DevOps Concepts

DevOps Concepts Content Coverage

Agile vs waterfall

- Agile Process model's
- Scrum

What is DevOps?

- Where did DevOps Come from?
- Timeline of DevOps
- How DevOps differ from traditional IT
- Build Automation

DevOps Principles & Culture

What is CI/CD/CT

- Continues Integration
- Pipeline
- CI Tools
- Continuous Delivery
- Continuous Deployment
- Continues Testing

Infrastructure as code

- What is infrastructure as code?
- Configuration management
- Tools

Containerization & Container orchestration

- containers vs virtual machines?
- Container Orchestration
- Container Orchestration tools
- Container Orchestration Platforms
- Monitoring

What is DevSecOps?

- Application Security
- Security Development Lifecycle
- DevOps to DevSecOps
- DevSecOps Advantage

DevOps Content Coverage

Outcomes

- High level understanding of DevOps
- Understanding of DevOps Technologies & Tools
- Understanding of DevSecOps

Cloud

Cloud Content Coverage

Cloud Computing Overview

- Basic Concepts

AWS Overview

- AWS Key Features
- AWS Global Infrastructure
- AWS Products and Services

Identity and Access Management(IAM)

- AWS Policies
- AWS Users
- IAM Groups
- IAM Roles

Elastic Compute Cloud(EC2)

- Overview of EC2
- Instance Types
- EBS and EFS
- Best Practices for EC2

Simple Storage Service(S3)

- Overview of S3
- How to get data into and out of S3
- S3 version control
- Secure and encrypt data on S3
- Best Practices for S3

Virtual Private Cloud(VPC)

- Overview of Virtual Private Cloud
- Public and Private VPC
- Public, Private and Elastic IP addresses
- Internet Gateway, Route Table
- Security Groups, Network ACLs
- VPC Best Practices

Cloud Content Coverage

High Availability/ Auto Scaling

- Overview of Availability Zones
- Overview of ELB
- Overview of API Gateway
- Overview of Route 53
- Auto Scaling
- Best Practices

Relational Database Service(RDS)

- Measure performance
- Assess limitations

Monitoring and Metrics

- Overview of CloudWatch and CloudTrail
- EC2, EBS Metrics
- Overview of AWS Billing
- AWS Trusted Advisor

Application Services

- Key Application Services
- Overview of SQS, SNS and Step Function
- Application Services best practices

Containerization in AWS

- What is Docker
- ECS, ECR and Fargate Basic Concepts

AWS Lambda

- What is serverless Computing
- Introduction of AWS Lambda
- Building a Lambda function from Scratch

Deployment Automation

- Terraform

Security

- Guardduty
- WAF
- KMS
- Data Encryption

AWS Best Practices

Cloud Content Coverage

Outcomes

- Introduction to Cloud
- Understanding of cloud concepts
- Understanding of cloud best practices

DevSecOps

DevSecOps Content Coverage

Introduction to DevOps

Revisiting Traditional SDLC

Need of DevOps

DevOps History and evolution

- How the idea was formed and how it is evolved from agile

Devops Culture

- DevOps vs. Traditional Silos
- DevOps –wall of confusion
- Downsides of traditional silos
- What change DevOps can bring

DevOps Practice

- The three way
- The Principles of flow
- The principles of feedback
- The Principles of Continual Learning and experimentation

Build Automation

- What is build automation
- What does a build Automation do
- What do we need build automation
- Example of Build Automation

Continuous Integration(CI)

- What is CI
- What does a CI do
- What do we need CI
- CI Examples

Continuous Delivery & Continuous Deployment

- What is CD/CD
- What does a CD/CD
- What do we need CD/CD
- CD/CD example

Configuration Management

- What is CM
- What does a CM
- What do we need CM
- CM example

Orchestration

- What is Orchestration

DevSecOps Content Coverage

Orchestration

- What is Orchestration
- What does a Orchestration
- What do we need Orchestration
- Orchestration example

Monitoring

- What is Monitoring
- What does Monitoring do
- What do we need Monitoring
- Monitoring Examples

Microservices

- What is Microservices
- What does a Microservices
- What do we need Microservices
- Microservices example

Virtualization

- Hypervisor
- Containerization

Virtual Box

- Creating Virtual box with different OS

Introduction to DevOps tools

- Why tools are important to achieve
- DevOps Practice

Version Control

- Bitbucket
- Github
- Gitlab

BitBucket

- Introduction to bitbucket
- Getting familiar with Bitbucket Interface

Bit Bucket Practice

- Uploading a demo project to bit bucket and sharing with peers

DevSecOps Content Coverage

Introduction to DevOps tools

- Continues Integration Tool
- DevOps Practice

Compare Build Automation tools

- Jenkins
- Bamboo

Bamboo

- Understanding Bamboo
- Introduction to bamboo
- Bamboo Terminology and concept

Bit Bucket Practice

- Create a simple plan in Bamboo to execute some lines of code on Script

Whats is DevSecOps

- Introduction to DevSecOps

Building the CI/CD Pipeline

- Different phases of CICD PipeLine CGP

Deployment

- Deployment it on an instance on a cloud
- Workflow of CI/CD

Blue/Green deployment strategy

- Understand how changes are made to an existing and deployed application without taking it down from the production

Compare the tools

- Jira
- Confluence
- CI Examples
- SonarQube
- VoraCode
- TerraForm
- Ansible

DevSecOps Content Coverage

Compare the tools

- SonarQube
- VoraCode

XL

- Deploy
- Release

XL Deploy and Release

- Deploy
- Release

Secure SDLC

- What is Secure SDLC

Secure SDLC Activities and Security Gates

- Security Requirements (Requirements)
- Threat Modelling (Design)

Static Analysis and Secure by Default (Implementation)

- What is Static Application Security Testing
- Static Analysis and Its challenges
- Embedding SAST tools into the pipelines
- Secrets scanning to prevent secret exposure in the code

Dynamic Analysis

- What is Dynamic Application Security Testing

Dynamic Analysis and Its challenges

- Security Requirements (Requirements)
- Threat Modelling (Design)

Static Analysis and Secure by Default (Implementation)

- What is Static Application Security Testing
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DevSecOps Content Coverage

OS Hardening, Web/Application Hardening

- OS Hardening, Web/Application Hardening (Deploy)

Security Monitoring/Compliance (Maintain)

- Security Requirements (Requirements)
- Threat Modelling (Design)

Embedding Security as part of CI/CD pipeline

- What is Static Application Security Testing
- Static Analysis and Its challenges

Hands-on:

- Create a CI/CD pipeline suitable for modern application SonarQube

Infrastructure as Code and Its Security

- What is Infrastructure as Code and its benefits

Platform + Infrastructure Definition + Configuration Management

- Discussion on Platform + Infrastructure Definition + Configuration Management

Difference between – Provisioning and Configuration management

- Compare– Provisioning and Configuration management

Introduction to Terraform

- Terraform features and benefits

Introduction to Ansible

- Introduction to Ansible

Ansible continues

- Benefits of Ansible
- Push and Pull based configuration management systems

Introduction to YAML

- Practice on YAML

Hands on Ansible

- Modules
- Tasks
- Roles
- Playbooks

DevSecOps Content Coverage

Introduction to containerization

- What are containers

Container tools example

- Dockers and rockets compare

Containerized Applications example

- Containerized Applications example

Benefits of containerization

- Benefits of containerization
- Example

Overheads of Containerization

- Overheads of Containerization

Hand-On Lab

- Containerization of application using Docker

Introduction to Orchestration

- Overheads of Containerization
- Dockers and rockets compare

Orchestration tool and benefits

- Dockers Swarm and Kubernetes

Kubernetes

- Demo

Kubernetes Practice

- Kubernetes Practice

DevSecOps Content Coverage

Outcomes

- Introduction to DevSecOps
- Understanding of the CICD pipeline
- Understanding of containers
- Introduction to XL Deploy
- Terraform Hands on
- Ansible Hands on